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Дисциплина: Компьютерные системы и сети

ОТЧЁТ
к лабораторной работе
на тему
«TRACEROUTE»

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Задание: необходимо разработать простейший аналог утилиты *traceroute*, изученной в рамках первой лабораторной работы. В качестве протокола передачи данных использовать *ICMP*.

Решение:

Для запуска утилиты вводим в консоль команду mytraceroute.exe:

```
C:\Windows\System32>mytraceroute.exe 8.8.8.8
```

Результат работы программы приведен на рисунке 1:

```
traceroute to 8.8.8.8 32 hops max

 1  8 ms  1 ms  2 ms  192.168.0.1
 2  9 ms  8 ms  8 ms  10.31.0.1
 3  7 ms  8 ms  10 ms  10.254.95.165
 4  7 ms  8 ms  7 ms  37.17.5.114
 5  7 ms  7 ms  8 ms  37.17.4.94
 6  7 ms  24 ms  7 ms  37.17.4.82
 7  7 ms  8 ms  8 ms  37.17.5.77
 8  8 ms  8 ms  7 ms  37.17.5.78
 9  8 ms  7 ms  8 ms  212.98.161.101
10  7 ms  8 ms  8 ms  185.11.76.41
11  7 ms  8 ms  7 ms  185.11.76.26
12  *     *     *
13  22 ms  16 ms  16 ms  216.239.58.135
14  16 ms  17 ms  16 ms  209.85.252.117
15  15 ms  16 ms  16 ms  8.8.8.8

Server was found!
```

Рисунок 1 – Результат работы программы

Для сравнения, на рисунке 2 приведен результат работы встроенной в Windows утилиты tracert:

```
C:\Windows\System32>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1     1 ms     1 ms     2 ms  192.168.0.1
  2    15 ms    13 ms     7 ms  10.31.0.1
  3    13 ms     5 ms     6 ms  10.254.95.165
  4    13 ms    12 ms    13 ms  37.17.5.114
  5    12 ms     8 ms     6 ms  reserved.a1.by [37.17.4.94]
  6     7 ms     6 ms     6 ms  reserved.a1.by [37.17.4.82]
  7     7 ms    14 ms    13 ms  37.17.5.77
  8    18 ms    14 ms    14 ms  37.17.5.78
  9    14 ms     7 ms     8 ms  msq.asr-km.bn.by [212.98.161.101]
 10    14 ms    15 ms     8 ms  185.11.76.41
 11    14 ms     6 ms     7 ms  185.11.76.26
 12     *        *        *     Request timed out.
 13    16 ms    15 ms    13 ms  216.239.58.135
 14    21 ms    13 ms    15 ms  209.85.252.117
 15    22 ms    14 ms    14 ms  dns.google [8.8.8.8]

Trace complete.
```

Рисунок 2 – Результат работы tracert

icmp

No.	Time	Source	Destination	Protocol	Length	Info
122	7.785615	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=111/28416, ttl=11 (no response found!)
123	7.793669	185.11.76.26	192.168.0.129	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
124	7.794523	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=112/28672, ttl=11 (no response found!)
125	7.802066	185.11.76.26	192.168.0.129	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
126	7.802861	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=120/30720, ttl=12 (no response found!)
153	10.806639	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=121/30976, ttl=12 (no response found!)
190	13.814497	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=122/31232, ttl=12 (no response found!)
231	16.830117	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=130/33280, ttl=13 (no response found!)
232	16.852341	216.239.58.135	192.168.0.129	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
233	16.852965	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=131/33536, ttl=13 (no response found!)
234	16.869137	216.239.58.135	192.168.0.129	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
235	16.869947	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=132/33792, ttl=13 (no response found!)
236	16.886056	216.239.58.135	192.168.0.129	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
237	16.886968	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=140/35840, ttl=14 (no response found!)
238	16.902686	209.85.252.117	192.168.0.129	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
239	16.903433	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=141/36096, ttl=14 (no response found!)
240	16.919916	209.85.252.117	192.168.0.129	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
241	16.921215	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=142/36352, ttl=14 (no response found!)
242	16.937217	209.85.252.117	192.168.0.129	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
243	16.938122	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=150/38400, ttl=15 (reply in 244)
244	16.952905	8.8.8.8	192.168.0.129	ICMP	74	Echo (ping) reply id=0x47fc, seq=150/38400, ttl=117 (request in 243)
245	16.953789	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=151/38656, ttl=15 (reply in 246)
246	16.969593	8.8.8.8	192.168.0.129	ICMP	74	Echo (ping) reply id=0x47fc, seq=151/38656, ttl=117 (request in 245)
247	16.970497	192.168.0.129	8.8.8.8	ICMP	74	Echo (ping) request id=0x47fc, seq=152/38912, ttl=15 (reply in 248)
248	16.986140	8.8.8.8	192.168.0.129	ICMP	74	Echo (ping) reply id=0x47fc, seq=152/38912, ttl=117 (request in 247)
1548	93.565463	192.168.0.129	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=8164/58399, ttl=1 (no response found!)

Рисунок 3 – Пакеты в Wireshark

```
  ▸ Time to Live: 1
    Protocol: ICMP (1)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.129
    Destination Address: 8.8.8.8
    [Stream index: 6]
▾ Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x90d5 [correct]
    [Checksum Status: Good]
    Identifier (BE): 18428 (0x47fc)
    Identifier (LE): 64583 (0xfc47)
    Sequence Number (BE): 11 (0x000b)
    Sequence Number (LE): 2816 (0x0b00)
  ▸ [No response seen]
  ▸ Data (32 bytes)
```

Рисунок 4 – ICMP-request с TTL 1

```
▾ Internet Control Message Protocol
  ▸ Type: Time-to-live exceeded (11)
    Code: 0 (Time to live exceeded in transit)
    Checksum: 0xf4ff [correct]
    [Checksum Status: Good]
    Unused: 00000000
  ▸ Internet Protocol Version 4, Src: 192.168.0.129, Dst: 8.8.8.8
  ▾ Internet Control Message Protocol
      Type: Echo (ping) request (8)
      Code: 0
      Checksum: 0x90d5 [unverified] [in ICMP error packet]
      [Checksum Status: Unverified]
      Identifier (BE): 18428 (0x47fc)
      Identifier (LE): 64583 (0xfc47)
      Sequence Number (BE): 11 (0x000b)
      Sequence Number (LE): 2816 (0x0b00)
    ▸ Data (32 bytes)
```

Рисунок 5 – ICMP Time-to-live exceeded

```
    Time to Live: 15
    Protocol: ICMP (1)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.129
    Destination Address: 8.8.8.8
    [Stream index: 6]
▾ Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x904a [correct]
    [Checksum Status: Good]
    Identifier (BE): 18428 (0x47fc)
    Identifier (LE): 64583 (0xfc47)
    Sequence Number (BE): 150 (0x0096)
    Sequence Number (LE): 38400 (0x9600)
    [Response frame: 244]
  ▸ Data (32 bytes)
```

Рисунок 6 – Конечный ICMP-request с TTL 15

```
    Time to Live: 117
    Protocol: ICMP (1)
    Header Checksum: 0x7488 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 8.8.8.8
    Destination Address: 192.168.0.129
    [Stream index: 6]
▾ Internet Control Message Protocol
    Type: Echo (ping) reply (0)
    Code: 0
    Checksum: 0x984a [correct]
    [Checksum Status: Good]
    Identifier (BE): 18428 (0x47fc)
    Identifier (LE): 64583 (0xfc47)
    Sequence Number (BE): 150 (0x0096)
    Sequence Number (LE): 38400 (0x9600)
    [Request frame: 243]
    [Response time: 14,783 ms]
  ▸ Data (32 bytes)
```

Рисунок 7 – ICMP-reply